

AUTONOMOUS DELIVERY BOT FOR NERC

Domino Delivery and Blue-Box Targeting System

Detailed Technical Report

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AUTONOMOUS DELIVERY BOT FOR NERC Domino Delivery and Blue-Box Targeting System

Figure 1. Final robot prototype. Department of Mechatronics Engineering College of E&ME;, NUST, Rawalpindi
Detailed technical documentation of the mechanical platform, FEA, navigation, colour sensing and autonomous domino delivery system.

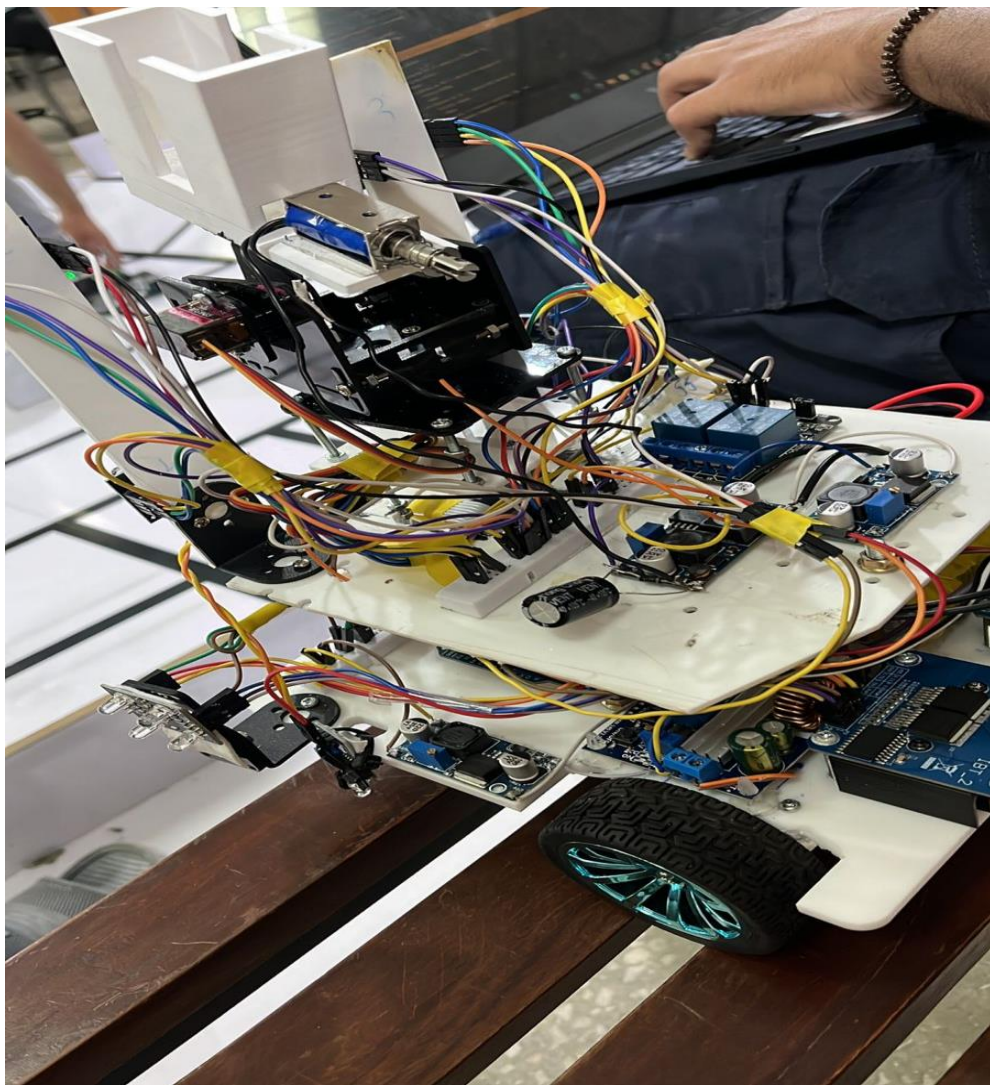


Figure from project analysis material — page 1

1. Project Overview and Mission

The completed robot is an autonomous delivery bot. Domino pieces are preloaded before the run. The robot autonomously navigates the course, uses colour sensing to identify the designated blue delivery box, aims its delivery mechanism, and shoots the domino into the blue box. The line-following system is therefore the robot's navigation method, not the main project purpose. The delivery function is the integration of navigation, colour recognition, aiming and launching. The supplied project reports document the mechanical design, machine-element calculations, FEA validation, QTR line sensing, IR junction detection, TCS3200 colour sensing, state-machine control and the two-servo/relay shooting mechanism. Ifilecitelturn13file1lturn12file1l

Mission flow

- Preload dominoes into the robot.
- Follow the programmed route using line sensing and junction detection.
- Reach the delivery region and identify the blue target/box with colour sensing.
- Aim the two-axis delivery mechanism.
- Launch/shoot the domino into the blue box.
- Continue or park according to the programmed route.

2. System Architecture and Requirements

Subsystem Main hardware Function Mechanical base 3 mm PMMA chassis, 6 mm steel shaft, 40 mm wheels
Structural platform Drive 2 × N20 12 V 150 RPM motors, differential drive Motion and turning Navigation QTR-8A + 2 external IR sensors Line following and junction detection Target sensing TCS3200 colour sensors Blue target identification Aiming Servo 1 + Servo 2 Vertical and horizontal aiming Delivery Relay-controlled firing actuator/launcher Domino delivery The mechanical report specifies a 160 × 120 × 3 mm PMMA chassis, 40 mm wheels, 6 mm steel shaft, N20 12 V 150 RPM motors, approximately 0.5 kg total mass and a minimum design FOS of 2. Ifilecitelturn13file1l The control report documents the QTR-8A array, external IR junction sensors, TCS3200 sensors, two servos and a relay-controlled firing system. Ifilecitelturn12file1lturn12file6l

3. Mechanical Design

The robot uses differential drive. The two independently driven wheels provide propulsion and steering by changing their relative speeds, while a castor provides passive support. The chassis carries the motors, shaft, sensors, electronics and delivery assembly. The compact PMMA chassis is 160 mm × 120 mm × 3 mm. The 6 mm steel shaft transfers torque to the 40 mm wheels. The QTR bracket places the line sensor near the floor for reliable reflectance measurements. Ifilecitelturn13file1l

Figure 2. Physical robot platform.

4. Mechanical Analysis

Differential-drive kinematics

Linear velocity: $v = (v_R + v_L) / 2$

Angular velocity: $\omega = (v_R - v_L) / L$

For $r = 0.02$ m and the documented motor angular speed of about 15.71 rad/s, the documented wheel speed is

approximately 0.314 m/s. Ifileciteltturn12file0l

Shaft torsion

$J = \pi d^4/32 = 127.23 \text{ mm}^4$; with $T = 80 \text{ N}\cdot\text{mm}$ and $r = 3 \text{ mm}$, the documented torsional stress is approximately 1.89 MPa. The reported shaft FOS is approximately 132.3. Ifileciteltturn13file5l

Fasteners and wheel

The M3 fastener analysis reports approximately 1.03 MPa shear stress and $\text{FOS} \approx 242.7$. The wheel analysis uses 8 N

vertical load, 2.5 N traction force and 80 N·mm torque; the reported wheel FEA stress is 1.07 MPa with FOS 46.7.

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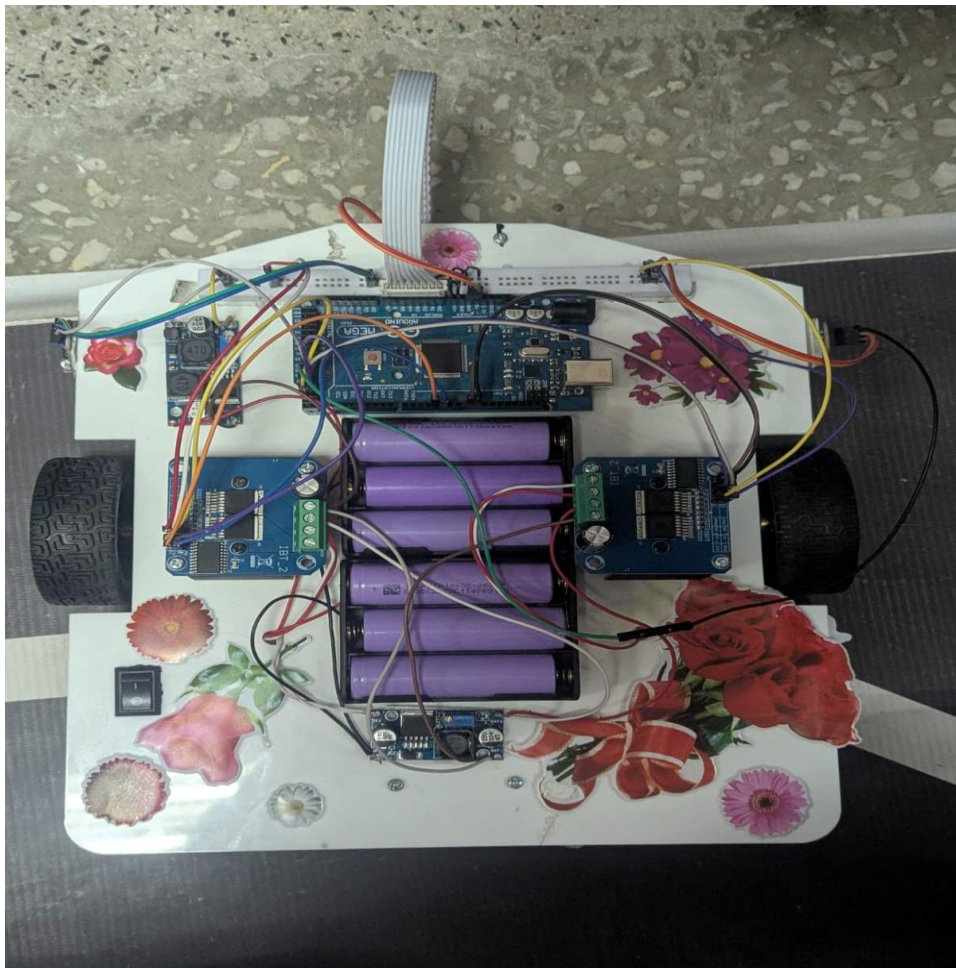


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5. FEA Analysis and Validation

FEA was used to validate the critical mechanical elements. The following figures are taken from the supplied project-analysis material.

Figure 3. Motor-bracket FEA stress result.

Figure 4. Motor-bracket FEA result. The motor bracket report gives maximum von Mises stress of 3.82 MPa and FOS 65.4. Ifilecitelturn13file9l The QTR bracket is documented with stress below 1 MPa and FOS above 250. Ifilecitelturn13file4l

Figure 5. QTR bracket FEA result.

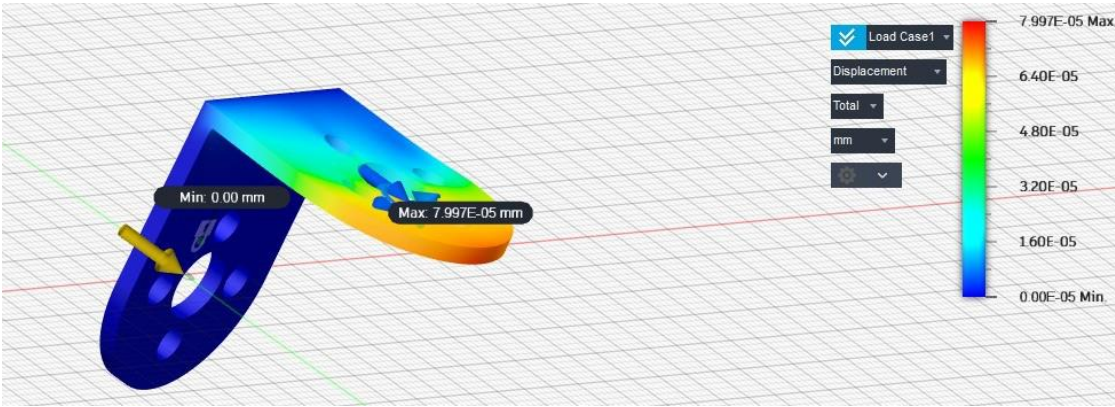


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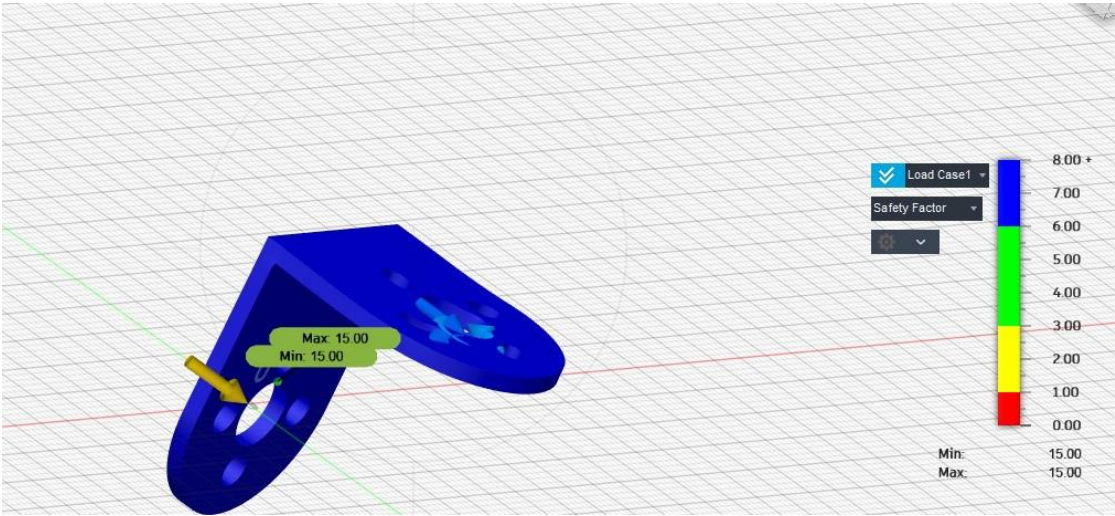


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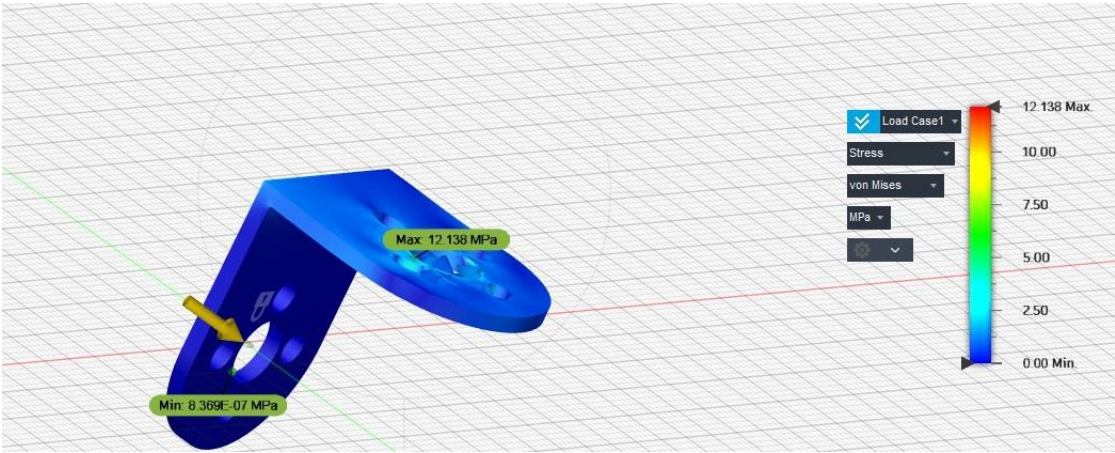
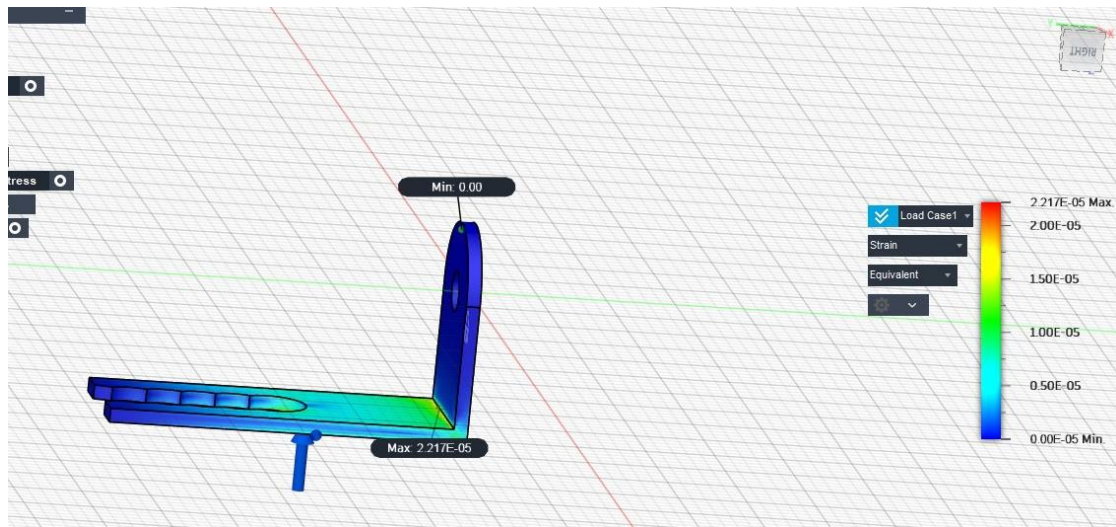


Figure 6. QTR bracket result/deformation view.



5.1 Wheel and Chassis FEA

The wheel FEA reports 1.07 MPa maximum von Mises stress and FOS 46.7. The chassis is the most deformation-sensitive component, with reported maximum FEA stress 3.55 MPa, PMMA yield strength 70 MPa, FOS

19.7 and maximum deformation 11.44 mm. Ifileciteturn13file2Itturn13file3I

Figure 7. Wheel FEA result.

Figure 8. Wheel analysis result.

Figure 9. Chassis displacement result.

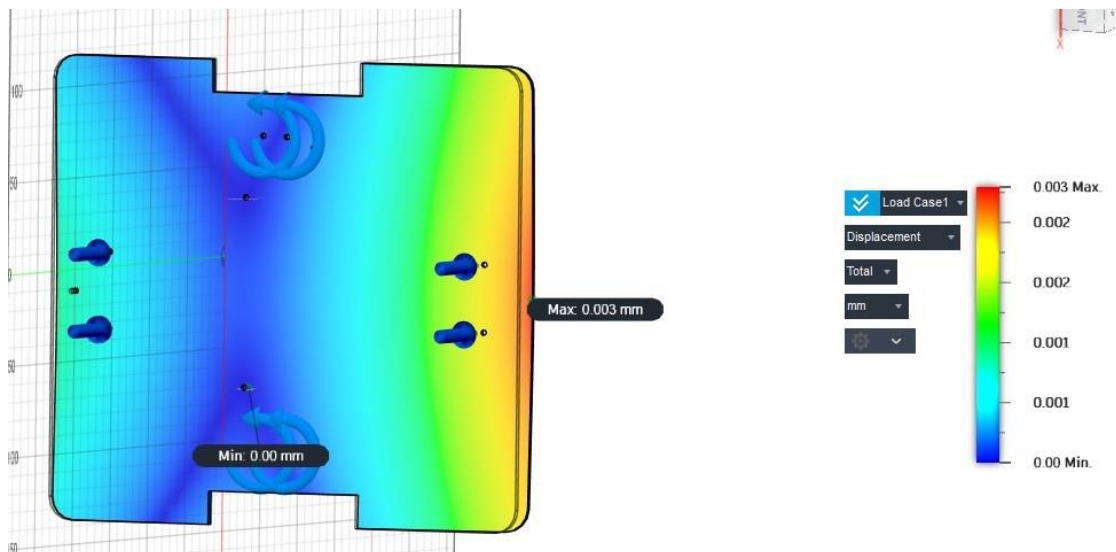




Figure 10. Chassis strain result.

Figure 11. Chassis stress/FEA view.

Figure 12. Chassis stress distribution.

Mechanical validation summary

Element Reported result FOS 6 mm steel shaft

1.89 MPa torsional stress

132.3 M3 fasteners

1.03 MPa shear stress

242.7 Motor bracket

3.82 MPa von Mises stress

65.4 QTR bracket < 1 MPa stress > 250

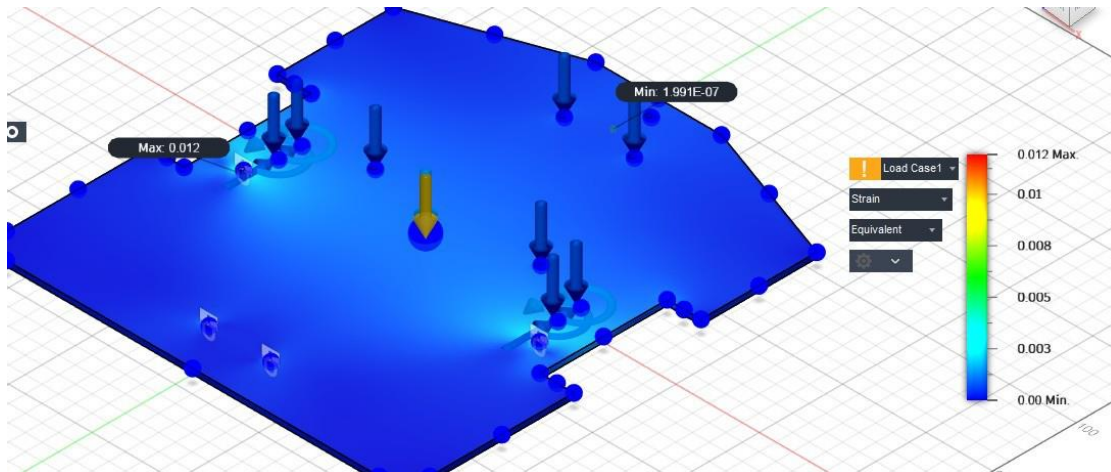


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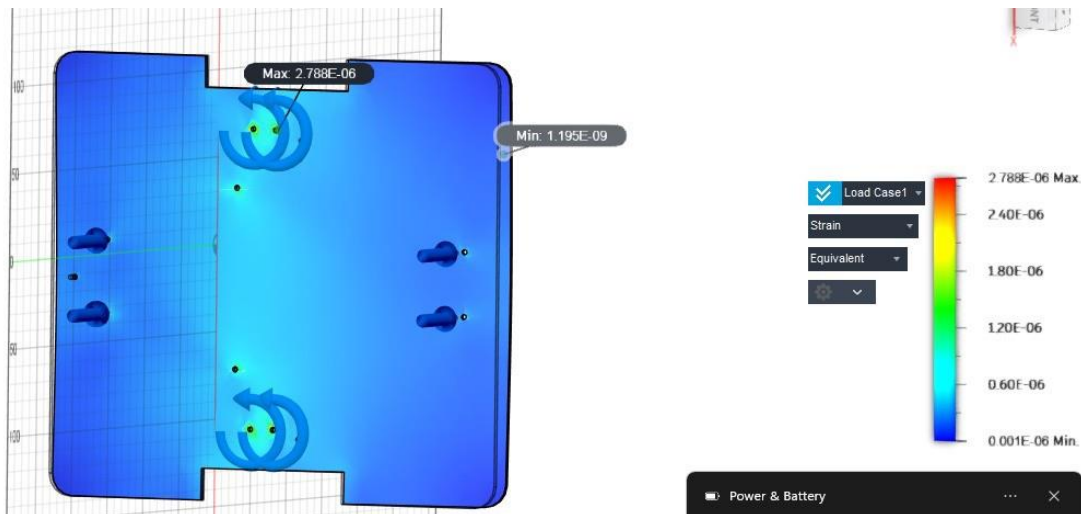
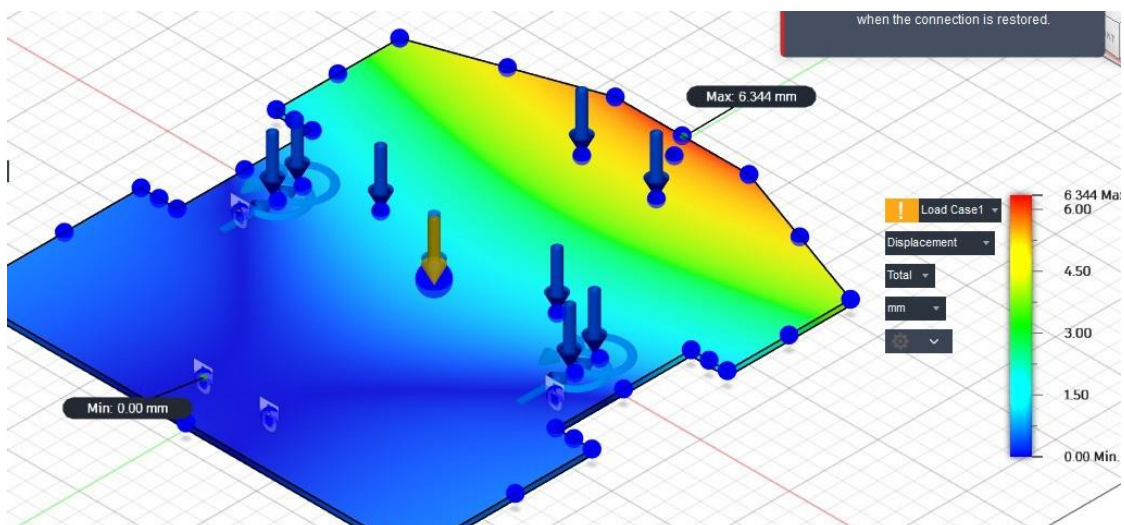


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Element Reported result FOS PC/ABS wheel

1.07 MPa von Mises stress

46.7 PMMA chassis

3.55 MPa von Mises stress

19.7 The source report states a minimum FOS of 2 and reports all six critical elements above that criterion. The chassis is identified as the lowest-FOS element and future stiffening/dynamic analysis is recommended.

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6. Navigation and Line Following

The QTR-8A array provides eight analogue reflectance channels. The documented implementation converts the sensor readings into a weighted line position from 0 to 7000, with 3500 as the nominal centre. Ifilecitelturn12file8l

Error = position – 3500

Correction = $K_p \times \text{error} + K_d \times (\text{error} - \text{lastError})$

Documented parameters are $K_p = 1$, $K_d = 3$, BaseSpeed = 90 PWM and MaxSpeed = 250 PWM, with correction

constrained to –50 to +50 PWM. Ifilecitelturn11file2l This controller keeps the robot on the route while the higher-level state machine handles junction decisions and the delivery sequence.

7. Junction Detection and Route Logic

Two external IR sensors are used to identify junctions. A junction is accepted when the robot is in FOLLOW state and the junction lock is inactive. The lock prevents the same physical junction from being counted repeatedly. QTR line reacquisition can also be used after special turns. Ifilecitelturn12file8l The state machine includes FOLLOW, timed turn states, delayed turns, reverse states, target scanning and PARKED. Junction number and stored blue-detection flags determine the next programmed action. Ifilecitelturn12file2lturn12file7l Thus the robot does not simply follow a line continuously: it follows the line until a decision point is detected, then executes the programmed route action before returning to navigation.

8. Colour Detection and Blue-Box Identification

Colour detection is central to the delivery mission. The TCS3200 measures red, green and blue filter outputs and the documented implementation uses ratios R/B, G/B and R/G to make the blue decision. The stored S1 blue thresholds are approximately R/B = 1.483, G/B = 1.441 and R/G = 1.029, with tolerance 0.2. Ifilecitelturn12file9l For target positions S2–S5, the sensor paths are enabled sequentially so the robot can determine which target position corresponds to the blue delivery box without simultaneous output conflict. Ifilecitelturn12file4l

Figure 13. TCS3200 colour sensor.

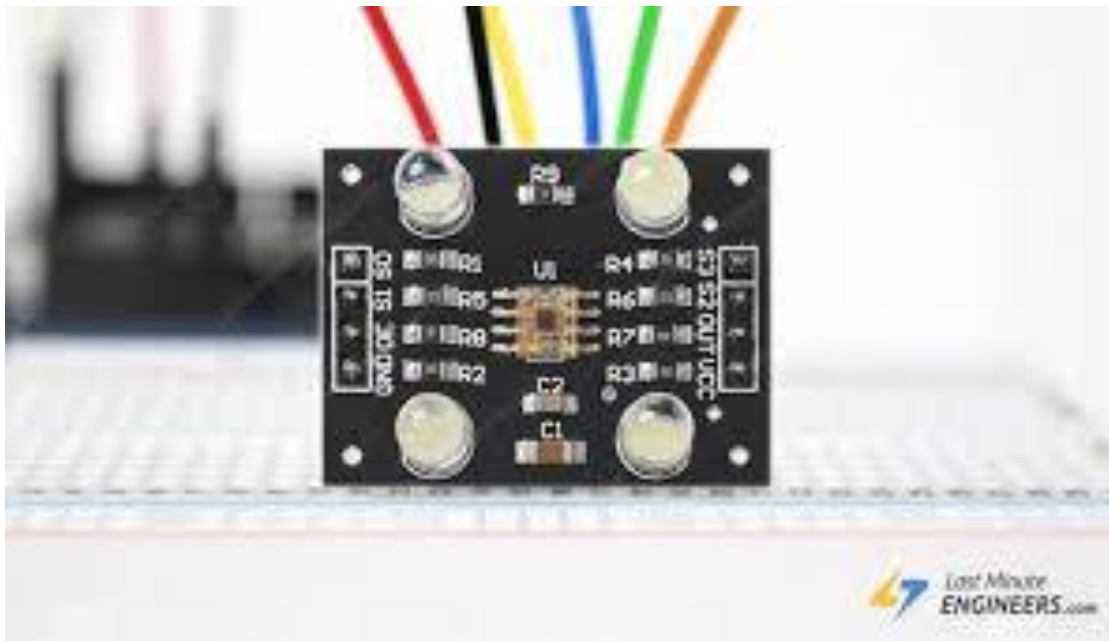


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9. Domino Delivery Mechanism

The robot carries preloaded domino pieces. Once the blue delivery box is identified, the delivery subsystem aims and launches the domino into the box. This is the defining task of the completed robot. The documented mechanism uses two servos: Servo 1 for vertical aiming and Servo 2 for horizontal/side aiming. A relay controls the firing actuator. The project material also documents a 3D-printed guide/support associated with the launcher. Ifilecitelturn12file6l

Figure 14. Launcher/delivery mechanism analysis image. The software moves the servos gradually toward the calibrated target angles, performs the firing pulse, and returns the servos toward neutral. The relay is documented as active-low and the firing pulse is approximately 100 ms. Ifilecitelturn12file3l

Delivery sequence

Stage Action Purpose 1 Preload dominoes Carry payload 2 Navigate Reach delivery region 3 Colour detect Identify blue box 4 Aim Align launcher 5 Fire Shoot domino into box 6 Reset Return mechanism toward neutral

10. Autonomous Mission Sequence

- Domino pieces are preloaded before starting.
- Robot enters FOLLOW and uses QTR feedback for navigation.
- IR sensors identify programmed junctions.
- State-machine logic chooses the route according to junction number and stored colour-detection information.
- At the delivery stage, TCS3200 sensing identifies the blue target box.
- Servo 1 and Servo 2 position the delivery mechanism.
- The relay activates the launcher and the domino is shot into the blue box.
- The mechanism returns toward neutral and the robot continues or parks according to the programmed state.

11. Final Integrated Robot

The final prototype integrates the mechanically analysed base, drive electronics, QTR navigation, IR junction detection, colour sensing and two-axis delivery mechanism into one autonomous platform.

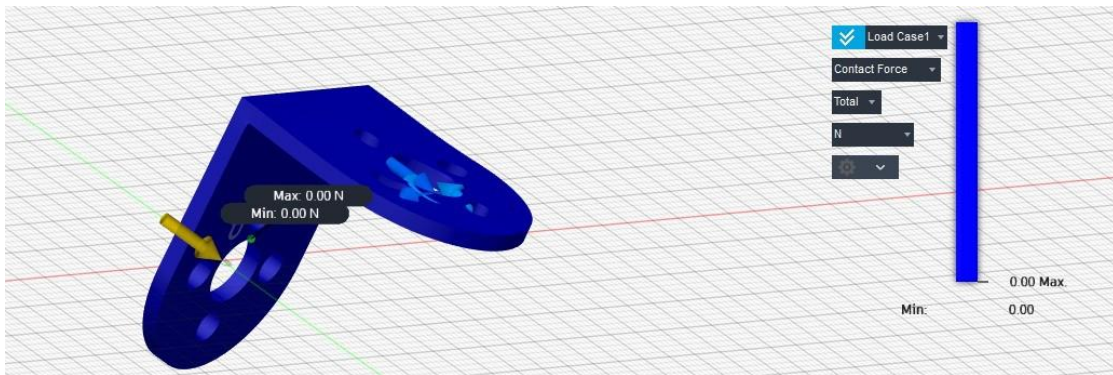


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Figure 15. Final integrated robot assembly.

Figure 16. Final prototype photograph. The project therefore demonstrates multiple engineering domains in one system: mechanical CAD and machine-element analysis, FEA, embedded control, sensor integration, autonomous navigation, colour recognition, servo positioning and payload delivery.

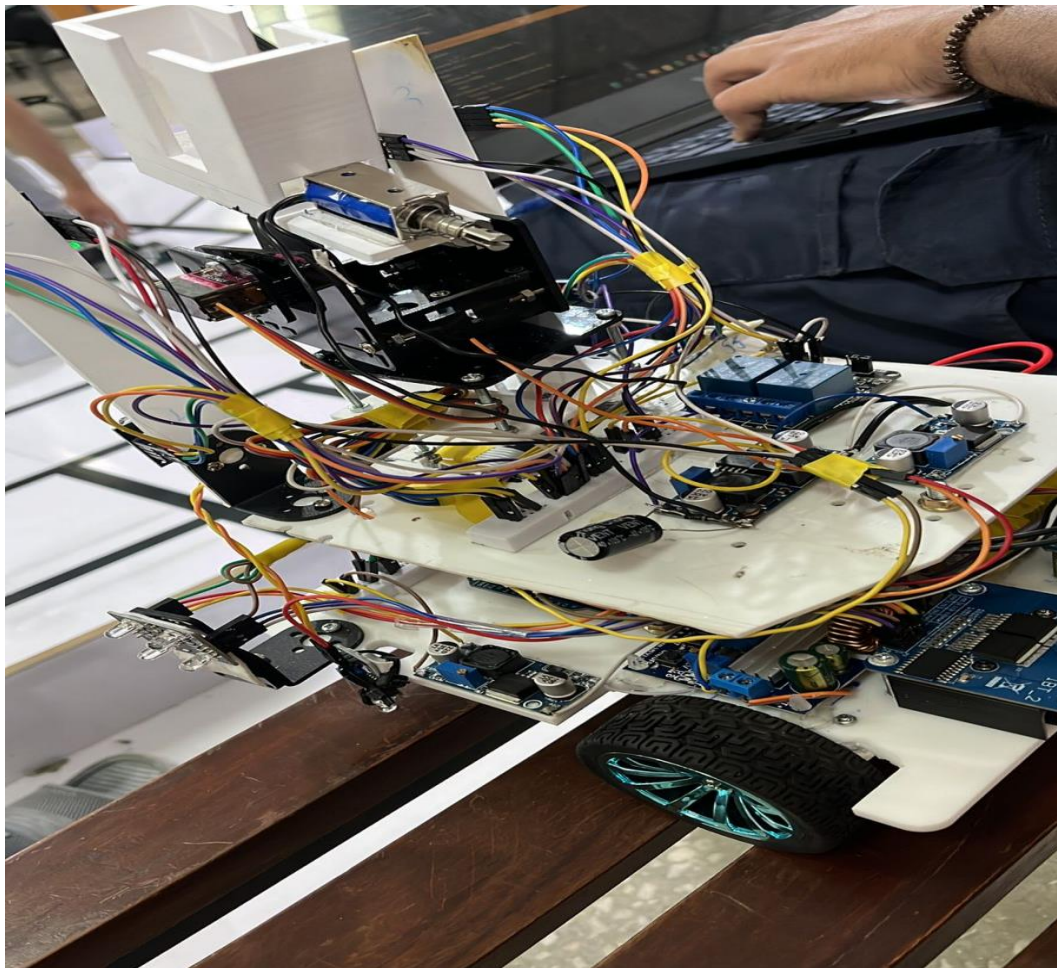


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12. Testing and Validation

Mechanical validation is supported by analytical calculations and FEA. The reported factors of safety exceed the stated minimum criterion. The control implementation includes several measures intended to improve repeatability:

- QTR calibration.
- Bounded PD correction.
- Junction locking.
- Line reacquisition after special turns.
- Colour-settling delay and averaged readings.
- Sequential target-sensor activation.
- IR ignore interval after special turns.
- Line-search timeout.
- Gradual servo movement.
- Neutral servo return after delivery.

13. Engineering Limitations and Future Improvements

The mechanical report identifies the chassis as the most deformation-sensitive component and recommends stiffening ribs, greater thickness around motor mounts and future dynamic analysis. For the delivery function, useful future measurements would include blue-box detection accuracy under different lighting, domino landing accuracy, repeatability of aiming angles, delivery success percentage, delivery time, battery current during firing and launcher alignment after repeated operation.

14. Conclusion

The project is best represented as an Autonomous Delivery Bot for NERC. Line following is the navigation method; colour sensing identifies the blue delivery box; and the two-servo/relay launcher performs the actual domino delivery. The engineering development covers the full chain from mechanical design and FEA through embedded sensing and control to autonomous target recognition and physical payload delivery. The detailed evidence in this report makes the repository suitable for demonstrating both the engineering analysis and the final working concept.